Cypress E2E

www.cypress.io

Prerequisites and Requirements

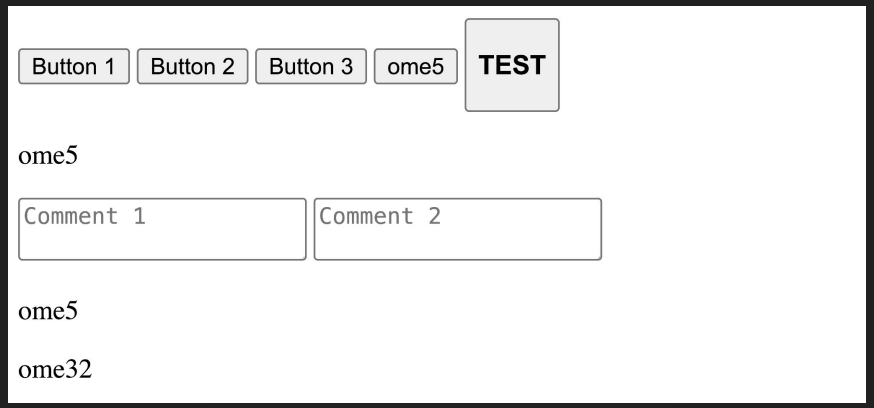
- Basic to intermediate HTML/CSS knowledge
- Basic to intermediate JavaScript or TypeScript knowledge
- Basic to intermediate web front-end knowledge

- Node.js 18.x
- Node.js 20.x and above
- npm



https://nodejs.org/en/download

Warming Up




```
1.    <div id="buttonDiv">
2.        <button class="class1" id="button1">Button 1</button>
3.        <button class="class2" id="button2">Button 2</button>
4.        <button class="class1" id="button3">Button 3</button>
5.        <button class="class2" id="button4">ome5</button>
6.        <button class="class2" id="button4">
7.            <h3>TEST</h3>
8.        </button>
9.    </div>
10.   <div id="div1">
11.       <p class="class1">ome5</p>
12.       <textarea class="myClass" placeholder="Comment 1"
      customField="name"></textarea>
13.       <textarea class="myClass" placeholder="Comment 2"
      customField="surname"></textarea>
14.   </div>
15.   <div id="div2">
16.       <p class="class1">ome5</p>
17.       <p class="class2">ome32</p>
18.   </div>
```

Setting up



```
npm install cypress@12.17.3 --save-dev
npm install typescript --save-dev
npm install @types/node @types/cypress --save-dev
```

tsconfig.json



```
{
  "compilerOptions": {
    "target": "es5",
    "lib": ["es5", "dom"],
    "types": ["cypress"]
  }
}
```

cypress/tsconfig.json



```
{
  "compilerOptions": {
    "baseUrl": "../node_modules",
    "types": ["cypress"]
  },
  "include": ["**/*.ts"]
}
```



```
npx cypress open
```

cypress.config.ts



```
testIsolation: false,
```

Query

cy.contains()

Note : cy.contains() gets the first element

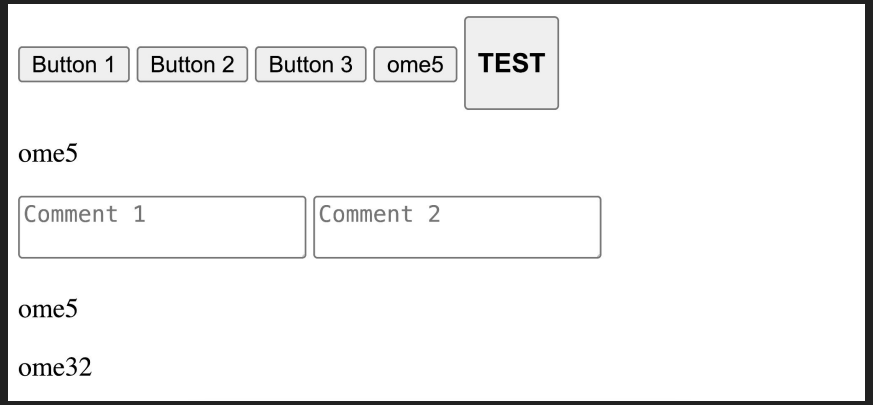
```
cy.contains('ome5');

cy.contains('Button');

cy.contains('p', 'ome5');

cy.contains('TEST');
```

https://docs.cypress.io/api/commands/contains




```
1.    <div id="buttonDiv">
2.        <button class="class1" id="button1">Button 1</button>
3.        <button class="class2" id="button2">Button 2</button>
4.        <button class="class1" id="button3">Button 3</button>
5.        <button class="class2" id="button4">ome5</button>
6.        <button class="class2" id="button4">
7.            <h3>TEST</h3>
8.        </button>
9.    </div>
10.   <div id="div1">
11.       <p class="class1">ome5</p>
12.       <textarea class="myClass" placeholder="Comment 1"
      customField="name"></textarea>
13.       <textarea class="myClass" placeholder="Comment 2"
      customField="surname"></textarea>
14.   </div>
15.   <div id="div2">
16.       <p class="class1">ome5</p>
17.       <p class="class2">ome32</p>
18.   </div>
```

Query

cy.get()

Note : cy.get() gets all element

```
cy.get('button');

cy.get('.class1');

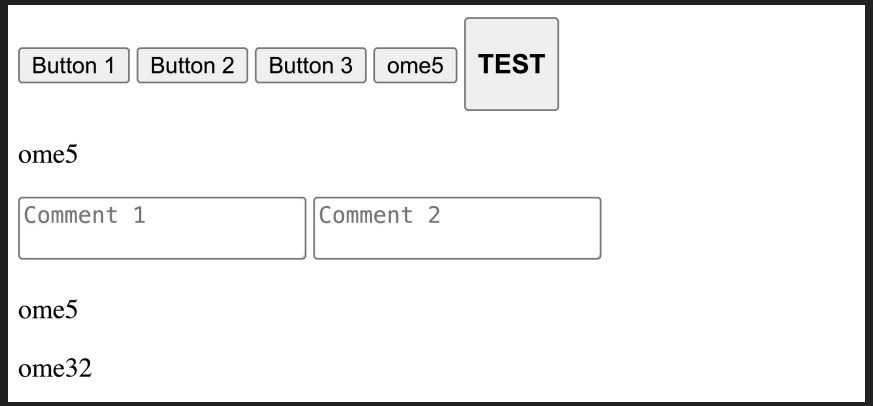
cy.get('#button2');

cy.get('#div2 > p');

cy.get('textarea[placeholder="Comment 1"]');

cy.get('.myClass[customField*="n"]');
```

https://docs.cypress.io/api/commands/get




```
1.    <div id="buttonDiv">
2.        <button class="class1" id="button1">Button 1</button>
3.        <button class="class2" id="button2">Button 2</button>
4.        <button class="class1" id="button3">Button 3</button>
5.        <button class="class2" id="button4">ome5</button>
6.        <button class="class2" id="button4">
7.            <h3>TEST</h3>
8.        </button>
9.    </div>
10.   <div id="div1">
11.       <p class="class1">ome5</p>
12.       <textarea class="myClass" placeholder="Comment 1"
      customField="name"></textarea>
13.       <textarea class="myClass" placeholder="Comment 2"
      customField="surname"></textarea>
14.   </div>
15.   <div id="div2">
16.       <p class="class1">ome5</p>
17.       <p class="class2">ome32</p>
18.   </div>
```

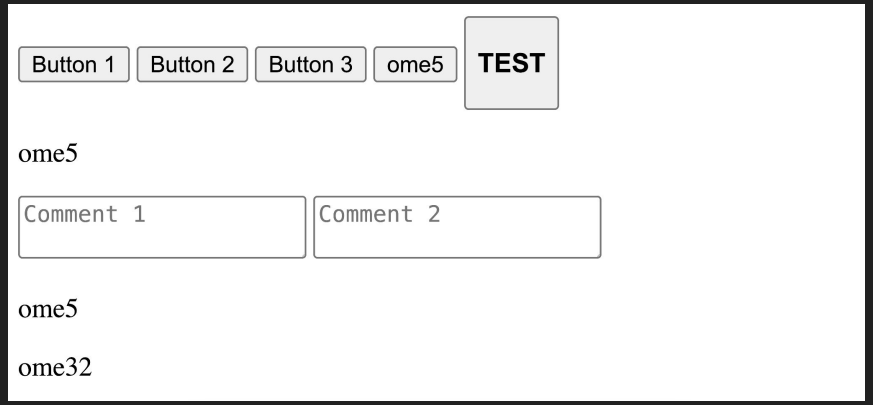
Query

Chaining query

```
cy.get('#div1').contains('ome5');

cy.get('#div2').get('.class2');

cy.get('#div2 > .class2');
```




```
1.    <div id="buttonDiv">
2.        <button class="class1" id="button1">Button 1</button>
3.        <button class="class2" id="button2">Button 2</button>
4.        <button class="class1" id="button3">Button 3</button>
5.        <button class="class2" id="button4">ome5</button>
6.        <button class="class2" id="button4">
7.            <h3>TEST</h3>
8.        </button>
9.    </div>
10.   <div id="div1">
11.       <p class="class1">ome5</p>
12.       <textarea class="myClass" placeholder="Comment 1"
      customField="name"></textarea>
13.       <textarea class="myClass" placeholder="Comment 2"
      customField="surname"></textarea>
14.   </div>
15.   <div id="div2">
16.       <p class="class1">ome5</p>
17.       <p class="class2">ome32</p>
18.   </div>
```

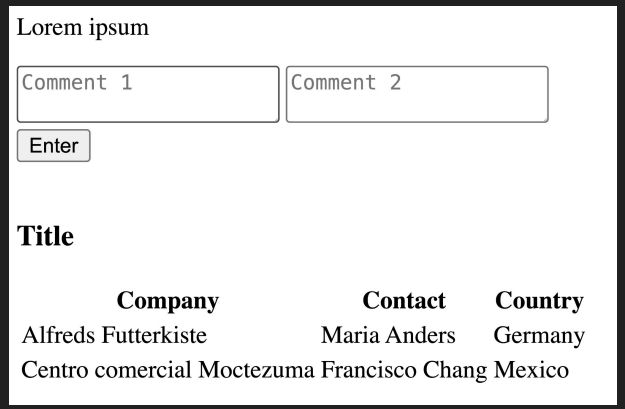
Query Traversal

```
.parent()

      cy.contains('Company').parent();

.parents()

      cy.contains('Company').parents();
```




```
1.    <div id="div1">
2.      <p class="class1">Lorem ipsum</p>
3.      <textarea class="myClass" placeholder="Comment 1"
      customField="name"></textarea>
4.      <textarea class="myClass" placeholder="Comment 2"
      customField="surname"></textarea><br>
5.      <button>Enter</button>
6.    </div><br>
7.    <div id="div2">
8.      <h3 class="myClass">Title</h3>
9.      <table>
10.       <tr id="header-row">
11.         <th>Company</th>
12.         <th>Contact</th>
13.         <th>Country</th>
14.       </tr>
15.       <tr>
16.         <td>Alfreds Futterkiste</td>
17.         <td>Maria Anders</td>
18.         <td>Germany</td>
19.       </tr>
20.       <tr>
21.         <td>Centro comercial Moctezuma</td>
22.         <td>Francisco Chang</td>
23.         <td>Mexico</td>
24.       </tr>
25.     </table>
26.   </div>
```

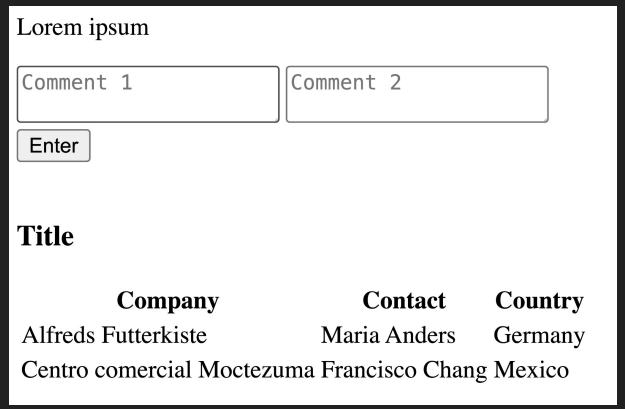
Query Traversal

```
.children()

      cy.get('#header-row').children();

.siblings()

      cy.contains('Contact').siblings();
```




```
1.    <div id="div1">
2.      <p class="class1">Lorem ipsum</p>
3.      <textarea class="myClass" placeholder="Comment 1"
      customField="name"></textarea>
4.      <textarea class="myClass" placeholder="Comment 2"
      customField="surname"></textarea><br>
5.      <button>Enter</button>
6.    </div><br>
7.    <div id="div2">
8.      <h3 class="myClass">Title</h3>
9.      <table>
10.       <tr id="header-row">
11.         <th>Company</th>
12.         <th>Contact</th>
13.         <th>Country</th>
14.       </tr>
15.       <tr>
16.         <td>Alfreds Futterkiste</td>
17.         <td>Maria Anders</td>
18.         <td>Germany</td>
19.       </tr>
20.       <tr>
21.         <td>Centro comercial Moctezuma</td>
22.         <td>Francisco Chang</td>
23.         <td>Mexico</td>
24.       </tr>
25.     </table>
26.   </div>
```

Query Traversal

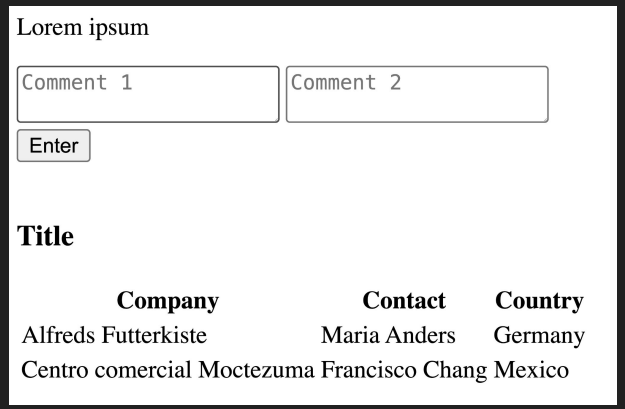
```
.filter()

      cy.get('.myClass');

      cy.get('#div1').children().filter('.myClass');

.eq()

      cy.get('tr').eq(2);
```




```
1.    <div id="div1">
2.      <p class="class1">Lorem ipsum</p>
3.      <textarea class="myClass" placeholder="Comment 1"
      customField="name"></textarea>
4.      <textarea class="myClass" placeholder="Comment 2"
      customField="surname"></textarea><br>
5.      <button>Enter</button>
6.    </div><br>
7.    <div id="div2">
8.      <h3 class="myClass">Title</h3>
9.      <table>
10.       <tr id="header-row">
11.         <th>Company</th>
12.         <th>Contact</th>
13.         <th>Country</th>
14.       </tr>
15.       <tr>
16.         <td>Alfreds Futterkiste</td>
17.         <td>Maria Anders</td>
18.         <td>Germany</td>
19.       </tr>
20.       <tr>
21.         <td>Centro comercial Moctezuma</td>
22.         <td>Francisco Chang</td>
23.         <td>Mexico</td>
24.       </tr>
25.     </table>
26.   </div>
```

More Query Traversal

```
.first()

.last()

.not()

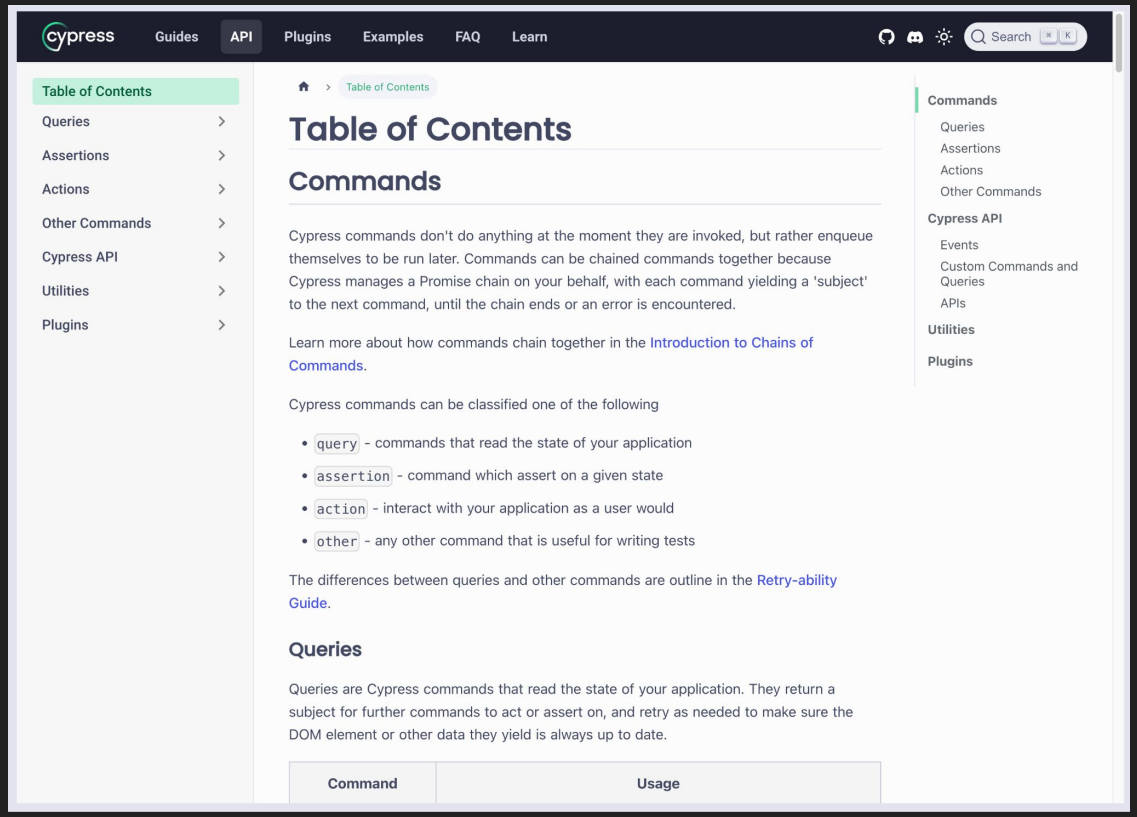
.prev()

.next()

.find()
```

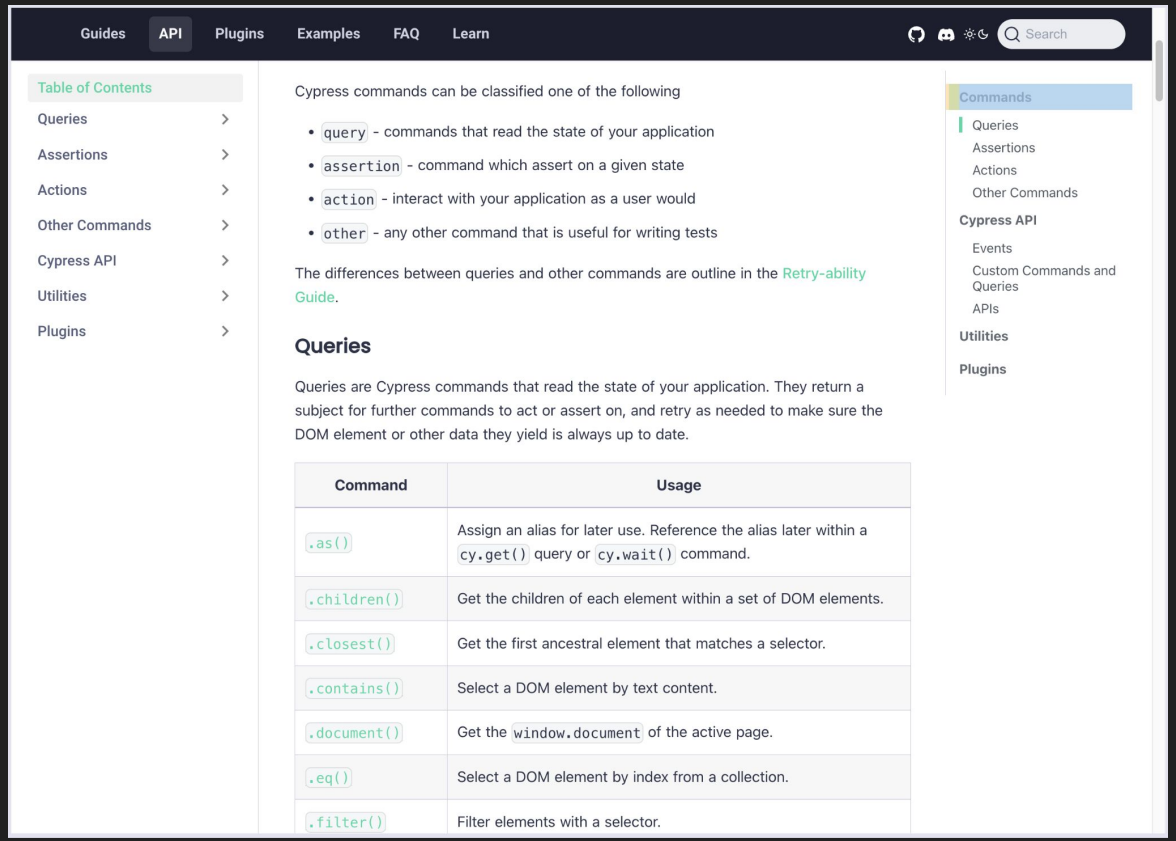
https://docs.cypress.io/api/table-of-contents#Queries

Exercise Query



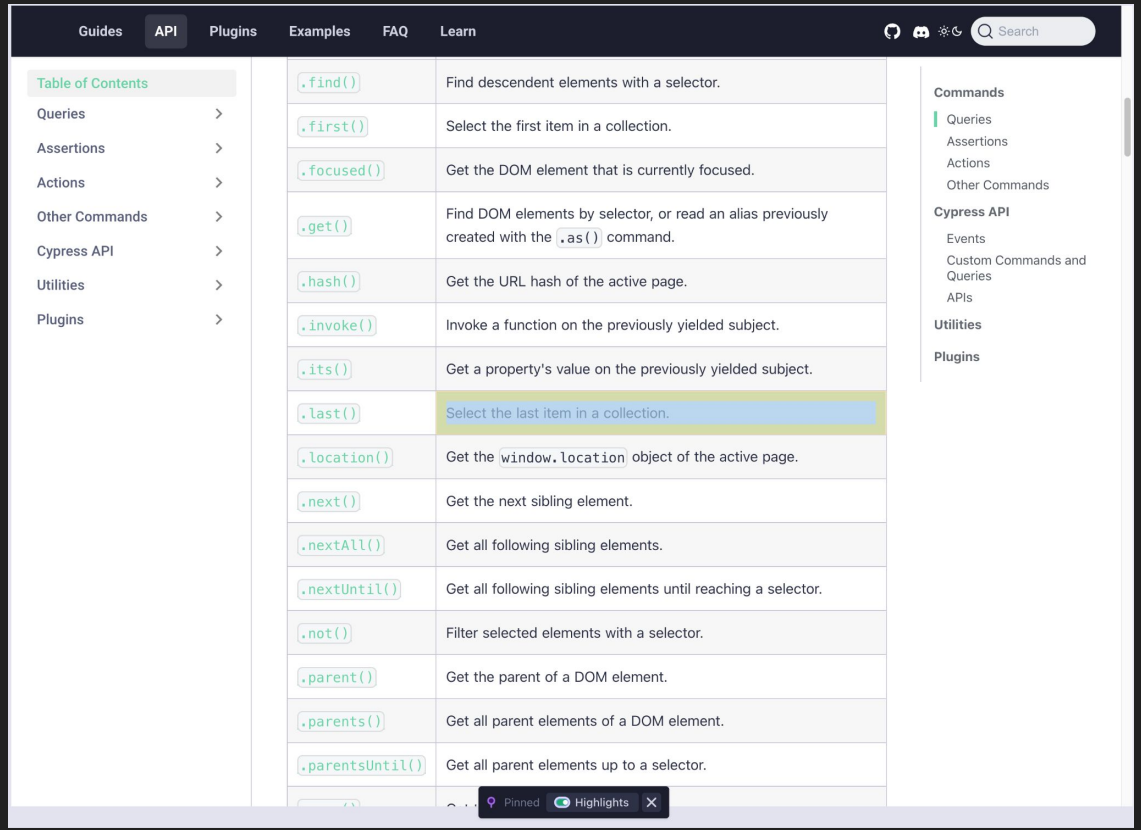
https://docs.cypress.io/api/table-of-contents

Exercise Query



Get highlighted element.

Exercise Query

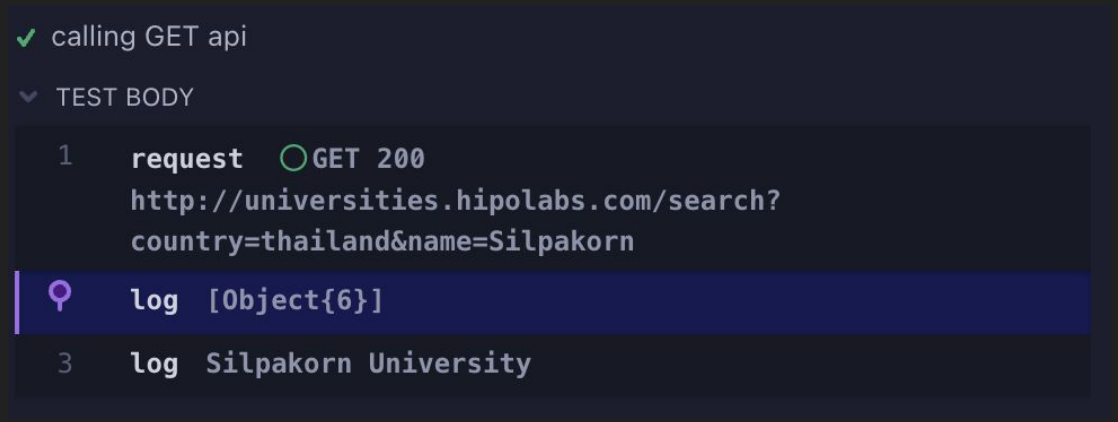


Get highlighted element without directly reference to it.

API Calling

Calling HTTP GET example

```
cy.request({
  method: 'GET',
  url: 'http://universities.hipolabs.com/search?country=thailand&name=Silpakorn',
}).then((response) => {
  let body = response.body;
  cy.log(body);
  cy.log(body[0].name)
});
```



API Calling (Fixture)

Using Fixture

```
let exampleReq : ExampleRequest;

describe('template spec', () => {

  before(() => {

    cy.fixture('example').then((data) => {

      exampleReq = data;

    });

  });

. . .
```

cypress/fixtures/example.json

```
{
  "name": "hahaha"
}
```

cypress/models/example_req.ts

```
export interface ExampleRequest  {
    name: string
}
```

API Calling

Calling HTTP POST example

```
cy.request({
  method: 'POST',
  url: 'https://httpbin.org/post',
  headers: { 'Content-Type': 'application/json' },
  body: exampleReq
}).then((response) => {
  let res = JSON.parse(response.body.data);
  cy.log(res.name);
});
```



Assert

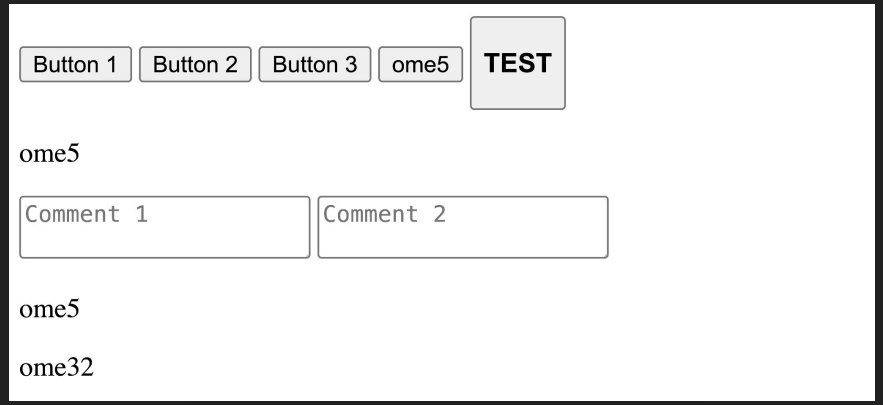
.should()

```
cy.get('button').should('have.length', 5);

cy.contains('ome32').should('exist');

cy.get('textarea[placeholder="Comment 1"]').type('Hello');

cy.get('textarea[placeholder="Comment 1"]').should('have.value', 'Hello');
```




```
<div id="buttonDiv">
    <button class="class1" id="button1">Button 1</button>
    <button class="class2" id="button2">Button 2</button>
    <button class="class1" id="button3">Button 3</button>
    <button class="class2" id="button4">ome5</button>
    <button class="class2" id="button4">
        <h3>TEST</h3>
    </button>
</div>
<div id="div1">
    <p class="class1">ome5</p>
    <textarea class="myClass" placeholder="Comment 1"
customField="name"></textarea>
    <textarea class="myClass" placeholder="Comment 2"
customField="surname"></textarea>
</div>
<div id="div2">
    <p class="class1">ome5</p>
    <p class="class2">ome32</p>
</div>
```

https://docs.cypress.io/guides/references/assertions

Assert

expect()

```
cy.request({
    method: 'POST',
    url: 'https://httpbin.org/post',
    headers: {'Content-Type': 'application/json'},
    body: {name: "hahaha"}
}).then((response) => {
    let res = JSON.parse(response.body.data);
    expect(res.name).to.equal('hahaha')
});
```

https://docs.cypress.io/guides/references/assertions

```
curl --location 'https://httpbin.org/post' \
--header 'Content-Type: application/json' \
--data '{name: "hahaha"}'
```


```
{
  "args": {},
  "data": "{name: \"hahaha\"}",
  "files": {},
  "form": {},
  "headers": {
    "Accept": "*/*",
    "Accept-Encoding": "gzip, deflate, br",
    "Cache-Control": "no-cache",
    "Content-Length": "16",
    "Content-Type": "application/json",
    "Host": "httpbin.org",
    "Postman-Token": "b37af52c-aad8-4e40-9115-5f266182938e",
    "User-Agent": "PostmanRuntime/7.36.0",
    "X-Amzn-Trace-Id":
"Root=1-657ed880-1537cc90247c9429436b53eb"
  },
  "json": null,
  "origin": "223.206.34.10",
  "url": "https://httpbin.org/post"
}
```

Actions

.click()

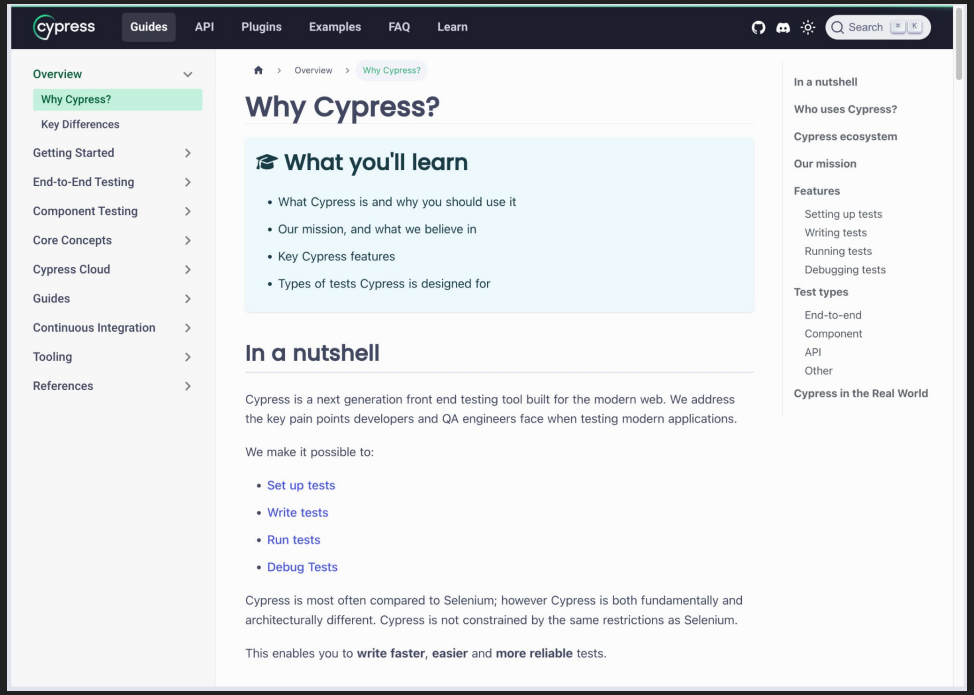
```
cy.get('#button1').click();
```

.type()

```
cy.get('textarea').type('Hello World!');
```

https://docs.cypress.io/guides/core-concepts/interacting-with-elements

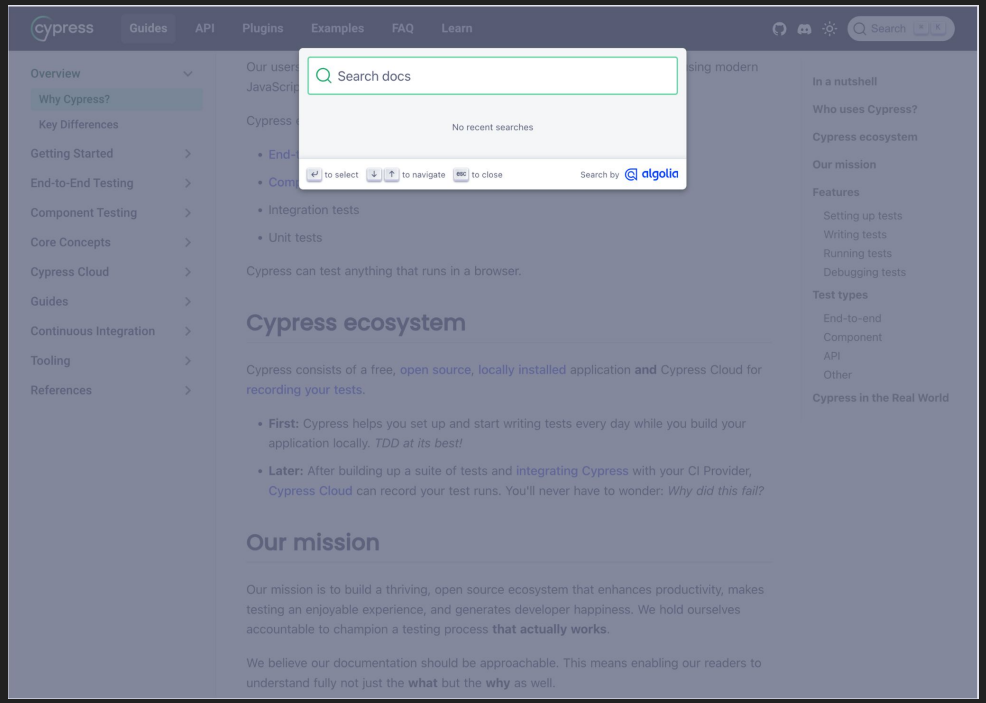
Actions in action.




```
it('enter site passes', () => {
    cy.visit('https://docs.cypress.io/')
  })

  it('Type Search', () => {
    cy.get('button[aria-label="Search"]').click()
    cy.get('input[placeholder="Search docs"]').type('click');
    cy.wait(500);
    cy.get('#docsearch-list').children().eq(1).click();
  })
```

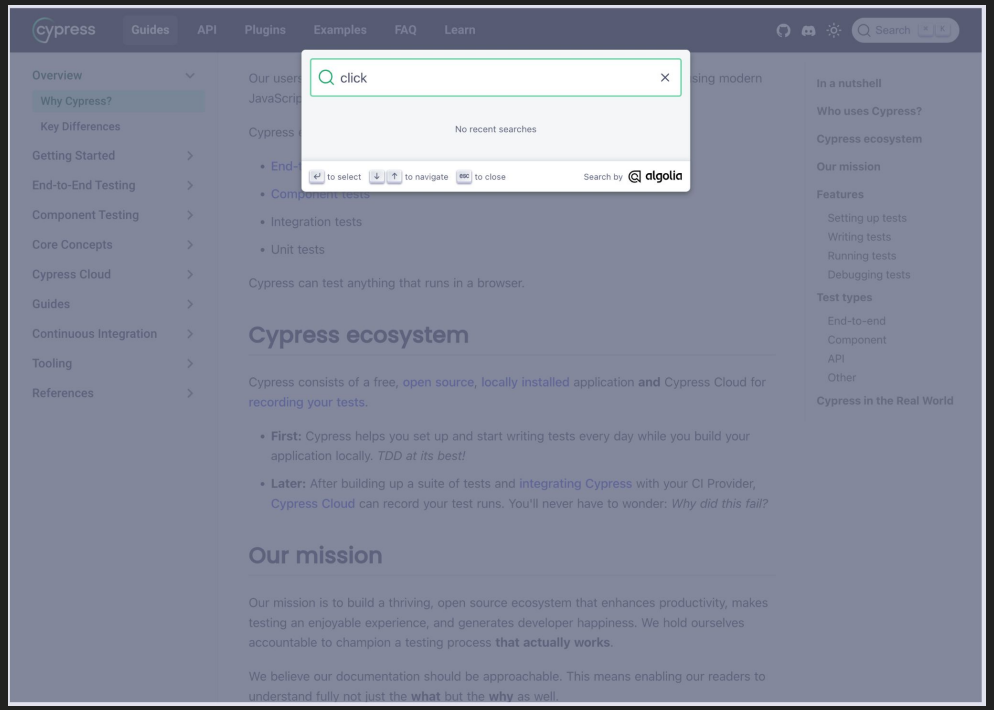
Actions in action.




```
it('enter site passes', () => {
    cy.visit('https://docs.cypress.io/')
  })

  it('Type Search', () => {
    cy.get('button[aria-label="Search"]').click()
    cy.get('input[placeholder="Search docs"]').type('click');
    cy.wait(500);
    cy.get('#docsearch-list').children().eq(1).click();
  })
```

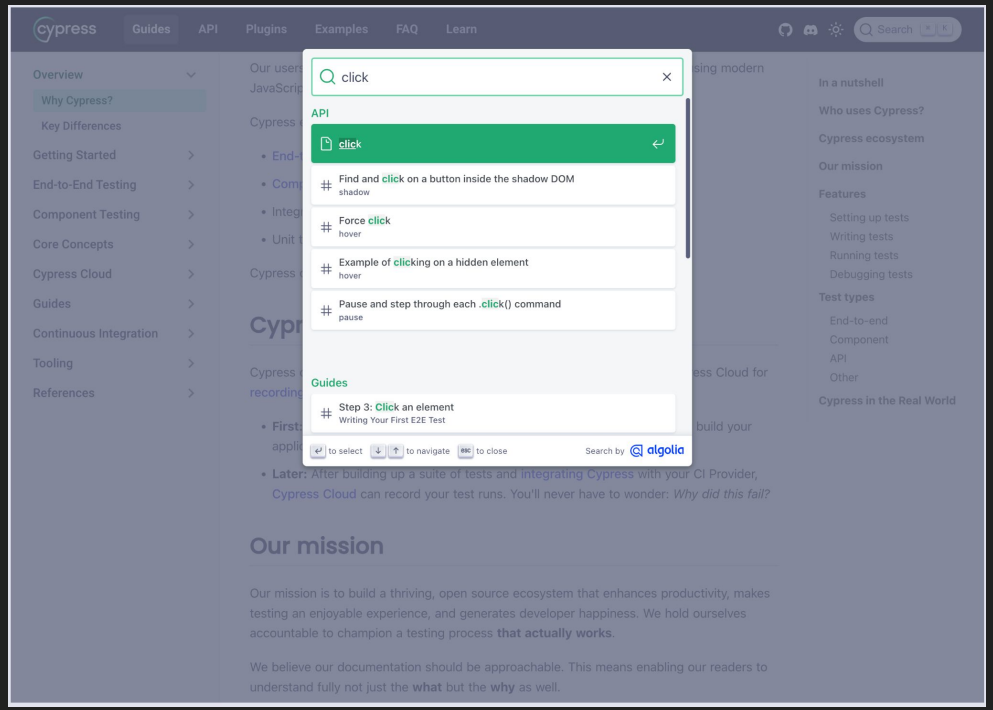
Actions in action.




```
it('enter site passes', () => {
    cy.visit('https://docs.cypress.io/')
  })

  it('Type Search', () => {
    cy.get('button[aria-label="Search"]').click()
    cy.get('input[placeholder="Search docs"]').type('click');
    cy.wait(500);
    cy.get('#docsearch-list').children().eq(1).click();
  })
```

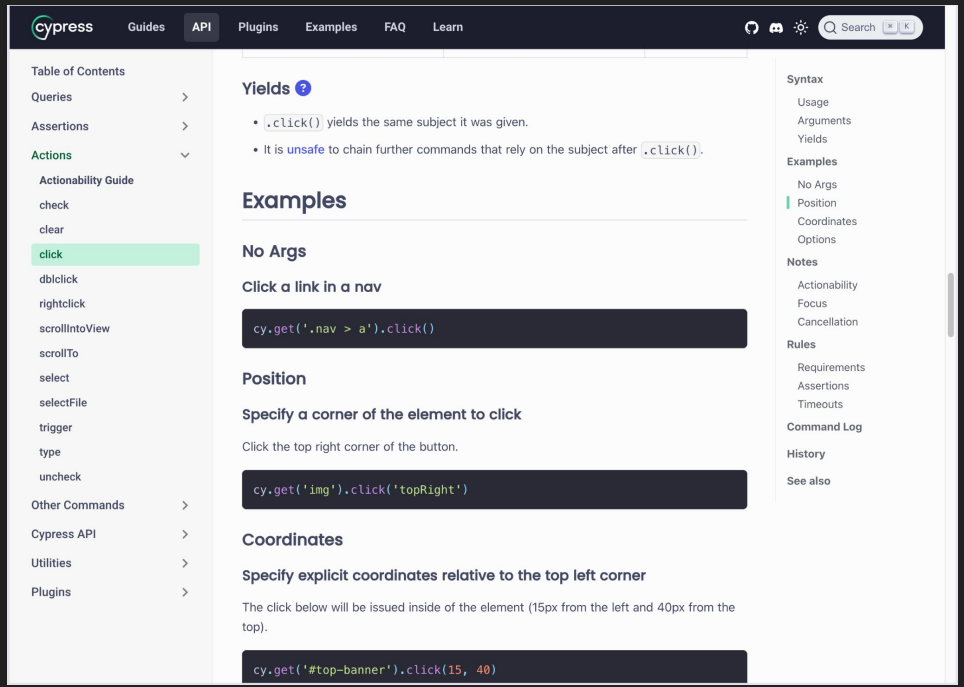
Actions in action.




```
it('enter site passes', () => {
    cy.visit('https://docs.cypress.io/')
  })

  it('Type Search', () => {
    cy.get('button[aria-label="Search"]').click()
    cy.get('input[placeholder="Search docs"]').type('click');
    cy.wait(500);
    cy.get('#docsearch-list').children().eq(1).click();
  })
```

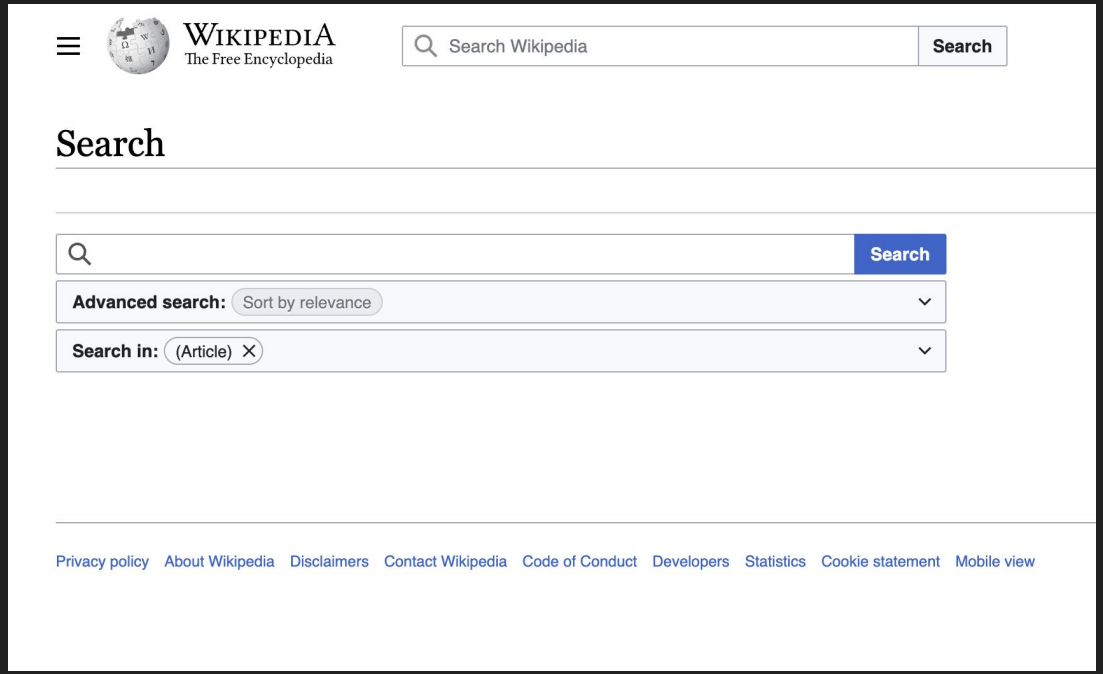
Actions in action.




```
it('enter site passes', () => {
    cy.visit('https://docs.cypress.io/')
  })

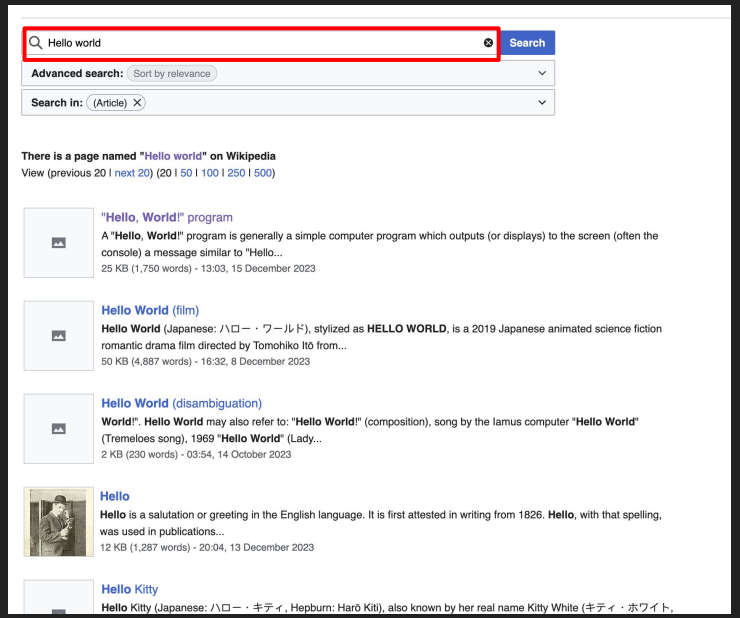
  it('Type Search', () => {
    cy.get('button[aria-label="Search"]').click()
    cy.get('input[placeholder="Search docs"]').type('click');
    cy.wait(500);
    cy.get('#docsearch-list').children().eq(1).click();
  })
```

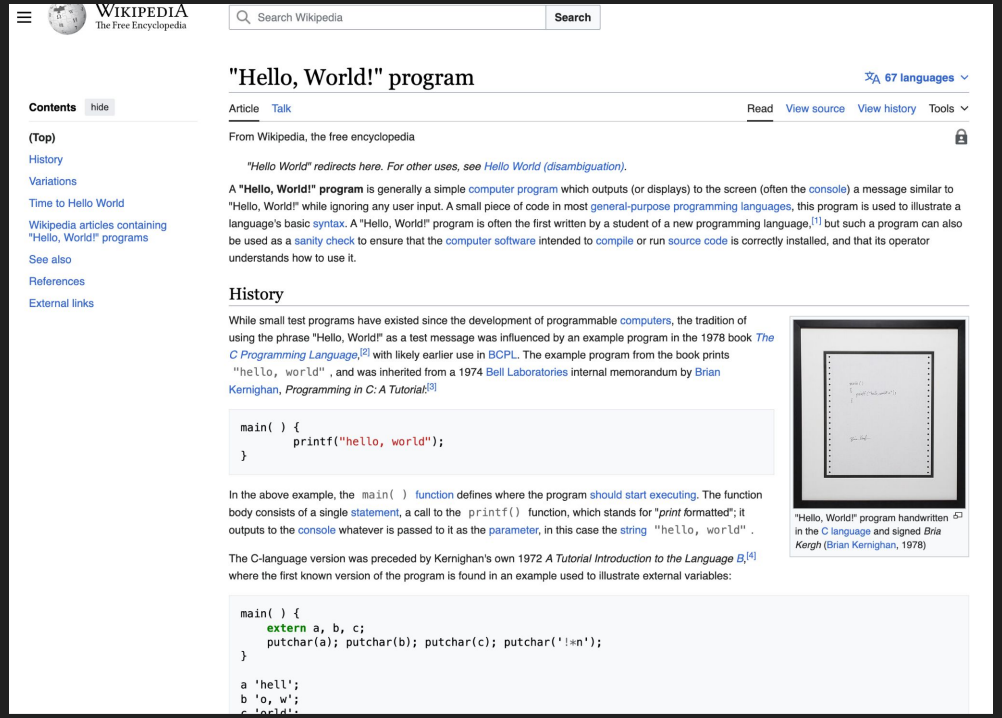
Exercise Action



https://en.wikipedia.org/w/index.php?search=&title=Special:Search

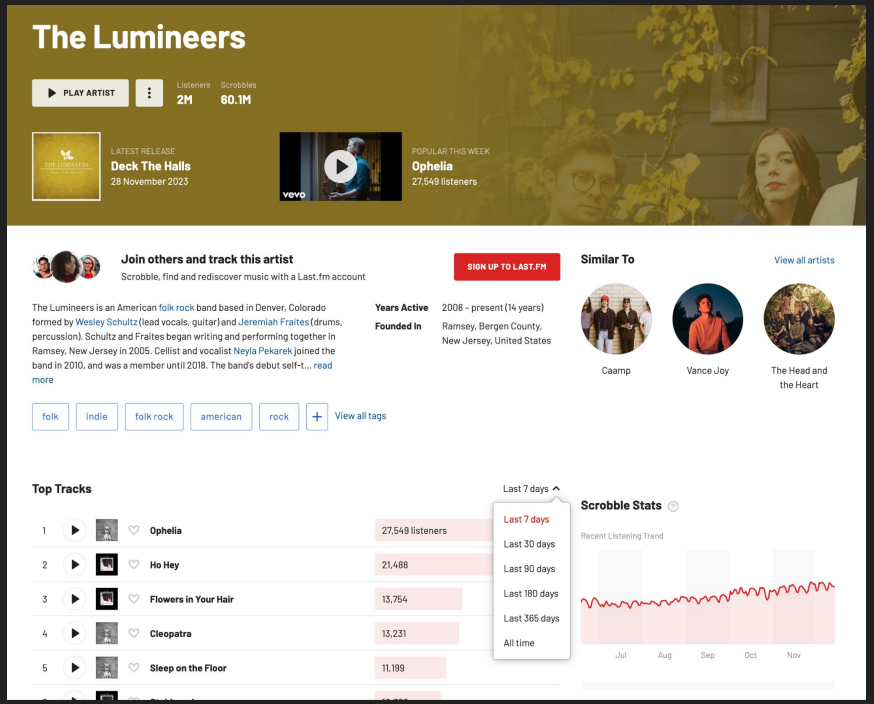
Exercise Action

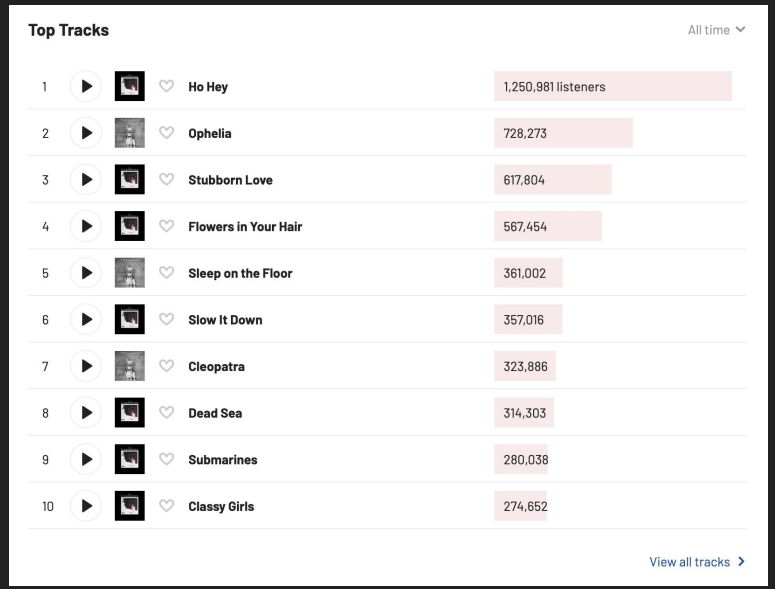




Search “Hello world” in the *Red Box* then go to the first result page.

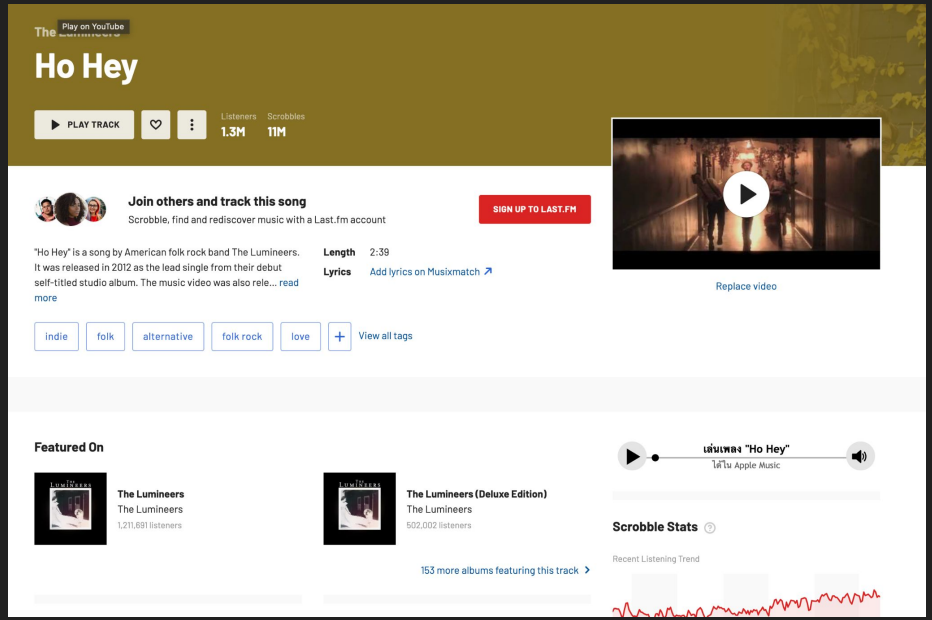
Intercept and Wait





https://www.last.fm/music/The+Lumineers

Intercept and Wait




```
cy.contains('button', 'Last 7 days').click({ force: true });
cy.intercept('GET', 'https://www.last.fm/music/The+Lumineers/+partial/tracks?top_tracks_date_preset*').as('datePreset');
cy.contains('a', 'All time').click();
cy.wait('@datePreset').then(() => {
  cy.get('tbody[data-playlisting-add-entries]').children().eq(0).children().contains('Ho Hey').click();
});
```

Custom Function

cypress/support/commands.ts

```
Cypress.Commands.add('typeTo', (placeholder :string, value: string) => {
    cy.get('textarea[placeholder="'+placeholder+'"]').type(value);
});
```

cypress/support/index.d.ts

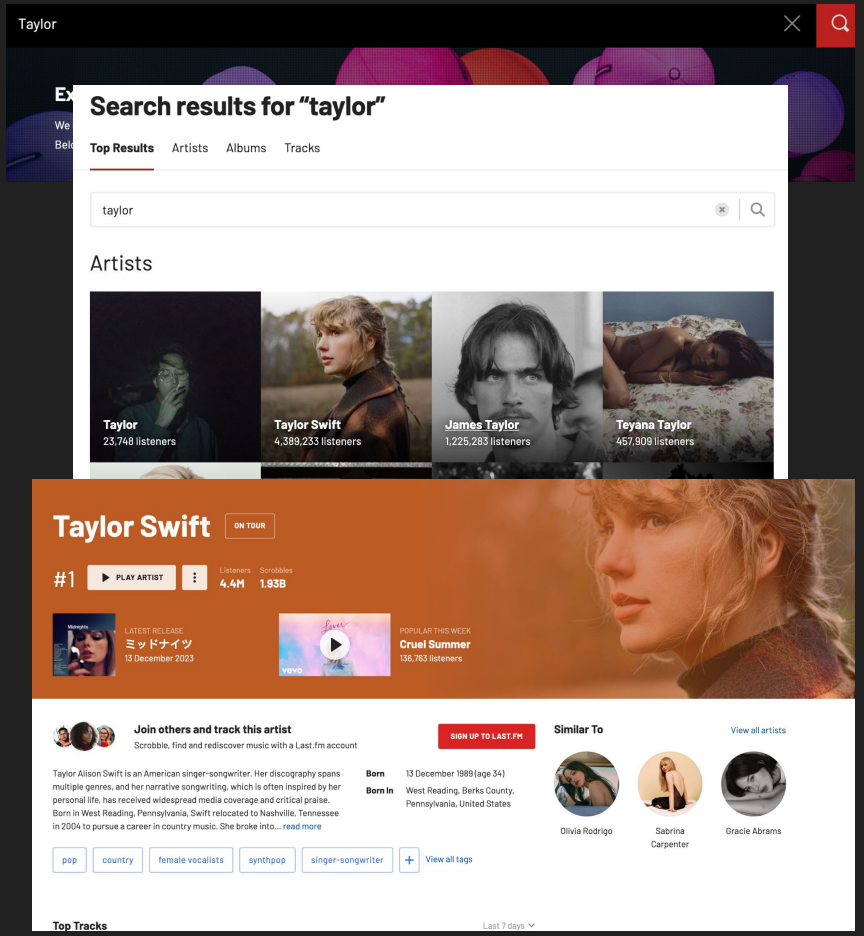
```
export { }
declare global {
    namespace Cypress {
        interface Chainable {
            typeTo(placeholder: string, value: string): Chainable<void>;
        }
    }
}
```

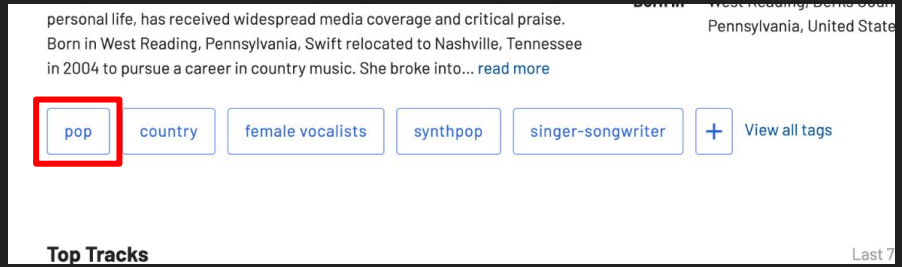
```
cy.typeTo('Comment 1', 'Hello');
```

Final Exercise

1) Goto https://www.last.fm
2) Search for “Taylor”
3) Go to Taylor Swift Page
4) Check of ‘poxp’ tag exist

Note : Considering if where to use intercept and wait or not at all






Question?